ABSTRACT

Development of Machine Learning-Based Anomaly Detection and Cyber Threat Prioritization System on Wazuh SIEM

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This thesis proposes a machine-learning–based anomaly detection artifact to reduce alert fatigue and improve triage effectiveness in a Wazuh SIEM environment. Instead of binary classification, the problem is formulated as ranking (Top-K/Top-p) so analysts can focus attention on the highest-value slice of events sized to daily capacity. The solution leverages a high-coverage feature spine with deterministic cleaning and schema canonicalization. Evaluation is temporally structured (train–validate–prospective test), with curated synthetic event injection during validation to provide operationally relevant ground truth. Results show that a neighbor-based approach (k-NN for anomaly scoring) is the most consistent under small triage budgets. At the operating point $p = 1\%$, the system attains precision@1% ≈ 0.68, recall@1% ≈ 0.23, lift@1% > 23× over the base rate, and FPR@1% ≈ 0.0033—indicating a high density of true positives within a small slice and controlled noise, with adequate day-to-day stability. An explainability layer via reason codes (contributing features, "normal neighbor" comparators, and context indicators) accelerates investigation handoff and strengthens prioritization accountability. The implementation is deployment-friendly through a frozen feature transformer and a headless service with a stable output contract, ready for integration with dashboards/orchestration. The main contributions include: (i) formulating detection as ranking tailored to triage; (ii) a feature spine resilient to attribute sparsity; (iii) a context-sensitive, explainable modeling approach; and (iv) an integration-ready implementation design. Limitations include incomplete network/geo/taxonomy signals and synthetic ground truth; future work involves expanding cleaned signals, labeling real incidents, and activating an end-to-end pipeline.

Kata kunci: anomaly detection, SIEM, Wazuh, ranking, k-NN, explainability, triage.